FedPromo: Federated Lightweight Proxy Models at the Edge for Fine-Grained Image Classification With Foundation Models

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***Abstract*—Federated learning (FL) is an established paradigm for training deep learning models on decentralized data, particularly relevant in Internet of Things (IoT) scenarios. However, as model sizes grow, conventional FL approaches require significant computational resources, which may not be feasible for resource-constrained IoT devices. We introduce FedPromo, a novel framework that enables efficient adaptation of classification heads trained in a federated way to large-scale foundation models (FMs) stored on a central server without requiring explicit data sharing. Instead of directly training the large model on client devices, FedPromo optimizes lightweight proxy models via FL, reducing computational overhead, energy consumption, and bandwidth usage while maintaining privacy. We evaluate our method in cross-domain fine-grained image classification via a two-stage process: server-side knowledge distillation (KD) aligns representations of a large-scale FM with those of a compact counterpart, then the compact model encoder is deployed to devices for local classifier learning. These classifiers are aggregated and transferred back to the FM, enabling learning of fine-grained representations without direct access to user data. Through novel regularization strategies, our framework enables decentralized multidomain learning, balancing performance, privacy, and resource efficiency for wide-scale IoT deployment. Extensive experiments on five image classification benchmarks and feature representation analysis demonstrate that FedPromo outperforms existing methods when employing mobile-targeted efficient architectures, making it suitable for IoT devices such as smart home devices, industrial sensors, and phones. Across domains, FedPromo enjoys average gains of 4.8% (9.1%) top-1 (top-5) accuracy on the clients and 7.5% (9.7%) on the server with respect to the best competitors.**

***Index Terms*—Federated learning (FL), foundation models (FMs), knowledge transfer.**

I. Introduction

In RECENT years, remarkable progress in visual understanding tasks has been achieved thanks to high-performance foundation models (FMs) based on complex transformer architectures [1], [2], [3], [4]. However, deploying such models on Internet of Things (IoT) and edge devices remains challenging due to their substantial computational demands [5], [6]. Despite improvements in edge devices, they still cannot efficiently run large-scale FMs [6]. This creates a critical gap between potential performance and practical deployment in IoT contexts where privacy, energy efficiency, and resource limitations are crucial. Real-world applications often generate valuable user data on user devices [7], but transferring it to centralized servers is often infeasible due to privacy issues, security concerns, and high communication costs. This raises a key question: how to transfer fine-grained private client information to a server-side FM that cannot access data on the devices, with minimal resources and energy consumption on the clients?

Federated learning (FL) [8], [9], [10], offers a natural solution to learning in decentralized settings that allows local model updates without exposing raw user data to a central server. However, conventional approaches assume that clients can locally train and execute the entire model [11], which becomes impractical for large-scale models that exceed the computational and energy constraints of the devices.

To address this issue, we introduce the cross-architecture federated knowledge transfer (CA-FKT) paradigm (Section III). This involves decentralized training of lightweight proxy models on fine-grained datasets, whose knowledge (in our specific case, a classifier head) is then integrated into a large-scale FM that cannot be deployed on IoT clients. Then, we propose a novel framework, FedPromo (see Fig. 1) to tackle it, verifying its effectiveness under several challenging cross-domain fine-grained image classification benchmarks. Our approach works in two stages, combining the strengths of cross-architectural knowledge distillation (KD) and FL.

1) *Server-Side Cross-Architectural KD:* Before deploying proxy models to IoT clients, we align the feature space of their lightweight CNN encoder (student) to that of a large-scale transformer-based model (teacher), ensuring that the small, compute-efficient model can approximate the embeddings of the larger model [12], [13].
2) *Federated Training of Proxy Models:* Proxy encoders are deployed to IoT devices and kept frozen to ensure consistency and minimize computation [14], [15], [16]. Local training focuses on a classifier head that is then

Received 13 May 2026; revised 10 June 2026; accepted 30 June 2026. Date of publication 6 July 2026; date of current version 26 August 2026. This work was supported in part by European Union under Italian National Recovery and Resilience Plan (NRRP) of NextGenerationEU, partnership on "Telecommunications of the Future" through the program "RESTART" under Grant PE00000001. *(Corresponding author: Pietro Zanuttigh.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/JIOT.2026.3710527, provided by the authors.

Digital Object Identifier 10.1109/JIOT.2026.3710527

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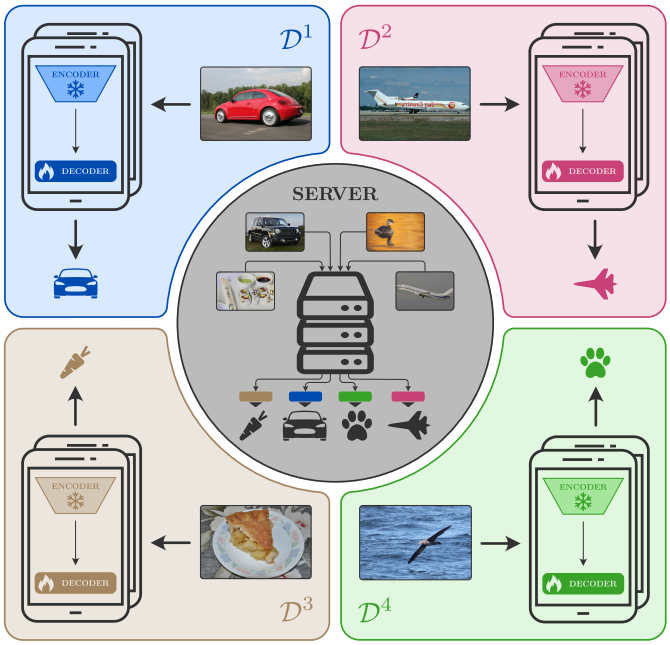


Fig. 1. FedPromo enables large-scale model adaptation via FL through compact proxy models. By combining cross-architectural distillation with specialized training strategies, it achieves private, decentralized, multidomain adaptation to data on resource-constrained clients.

aggregated at the server side and transferred to the FM, enabling multidomain learning without data access.

By freezing the encoder and training only the classifier, we achieve three objectives.

1) *Personalization:* Clients adapt their classifiers to local data domains while benefiting from the knowledge of the FM.
2) *Efficient Knowledge Aggregation:* The server integrates classifiers from multiple devices, facilitating knowledge transfer without access to private data.
3) *Resource Efficiency:* Minimizing computational load and energy consumption on client devices while reducing bandwidth requirements during federated training.

We introduce novel regularization strategies that preserve activations of classes not present on the selected client and reduce common activations between similar classes. Since the feature spaces of student and teacher models were aligned during pretraining, the aggregated classifier can be seamlessly plugged into the FM. This innovative approach enables adaptation of a large-scale model to specific IoT domains without direct access to training data, aligning with principles of modular deep learning by promoting composability, reusability, and personalization for privacy-preserving AI [17].

Summarizing, our contributions are as follows.

1) We introduce a novel paradigm, CA-FKT, for FL across heterogeneous model architectures in IoT environments.
2) We propose FedPromo, a scalable framework that leverages lightweight proxy models for bringing knowledge learned in a federated setting to a large-scale model, reducing energy consumption and bandwidth requirements.
3) We integrate cross-architectural KD with federated optimization to address key challenges in personalization, generalization, and efficiency for IoT.
4) We demonstrate the effectiveness of FedPromo in five cross-domain long-tailed classification benchmarks, achieving SOTA privacy-preserving adaptation for IoT scenarios.

The proposed approach has wide-ranging applications in IoT scenarios, including smart home systems, industrial IoT, and personal smartphones, enabling personalized fine-grain classification and adaptive quality control while maintaining user privacy and minimizing energy consumption.

II. Related Works

A. Federated Learning

FL enables decentralized learning without sharing private data, which is crucial for many vision tasks [10], [18], [19]. The first FL approach, i.e., FedAvg [11], is effective in simple scenarios, but struggles with non-IID distributions. FedProx [20] adds a proximal term to the local objectives to limit the impact of local updates, reducing clients' drift. SCAFFOLD [21] estimates client drift by comparing server and client update directions and uses it for correction. FedMargin [22] emphasizes the role of learned representations in model aggregation, while MOON [23] improves local training in non-IID settings through model-based contrastive learning. In IoT devices, FL tends to focus on efficiency [24] or on personalization, often considering approaches working at head-level [25], [26], [27].

B. Foundation Models

FMs have advanced multiple domains, beginning with natural language processing (*e.g.*, BERT [28]) and extending to computer vision. CLIP [1] marks a key development in vision-language models (VLMs), employing contrastive learning to align image-text representations for strong zero-shot transfer. DINO [29] uses self-distillation to learn semantic object structures without supervision, improving representation learning for downstream vision tasks. The rapid development of large-scale pretraining has resulted in FMs that can act as effective feature extractors for a variety of downstream tasks and domains. Ostapenko et al. [30] explored the efficacy of pretrained vision models as a foundation for downstream continual learning. FMs are also starting to gain interest in the context of FL [31]. Liu et al. [32] proposed fine-tuning a pretrained SAM [2] on each client and then introduce another approach of fine-tuning the parameters of the adapters and decoder of a medical SAM adapter. In this work, we improve an FM in a scalable, modular, and privacy-preserving way, using FL on remote devices.

C. Pretraining in FL

Traditional FL settings [11] limit the role of the server as a sole aggregator. However, given its computational capabilities, it is advantageous to fully exploit it for improved efficiency and performance. For this reason, recent works have begun

to explore pretraining steps at server-side before initiating local training on devices [10]. Pretraining allows us to better address data heterogeneity, allowing longer local training and reducing communication costs [33], [34]. Beyond improving communication efficiency, pretraining enables more realistic FL scenarios by addressing tasks where client-side annotations are impractical. For example, pretraining on synthetically generated supervised data facilitates unsupervised client-side learning for challenging vision tasks such as semantic segmentation [35].

D. Knowledge Distillation

Originally proposed to transfer knowledge from large to small models [36], KD is widely used in FL to address model and data heterogeneity [37]. Model-agnostic KD [38], [39], [40], [41] improves FL without architectural limits, while data-agnostic KD [42], [43], [44], [45] boosts robustness under non-IID data. In contrast, we integrate KD into pretraining, aligning lightweight models with the server before FL. This allows for transmitting only a small decoder attached to client and server models, reducing load.

III. Problem Setting

In this section, we introduce the CA-FKT task: as in conventional FL settings, we assume a central server s and a set of clients $\mathcal{K} = \{k\}_{k=1}^{K}$. We assume that the client set $\mathcal{K}$ is divided into G groups $\mathcal{K}^i, i = 1, \ldots, G$ such that $\mathcal{K} = \bigcup_i \mathcal{K}^i$ and $\mathcal{K}^i \cap \mathcal{K}^j = \varnothing \quad \forall i \neq j$. Each group $\mathcal{K}^i$ learns a model M^i from a private dataset $\mathcal{D}^i$. For ease of notation, we focus on a single domain and omit the index i in the following.

The FL process follows the typical communication scheme: clients perform local training, send their updated models to the server, which aggregates them, and redistributes the updated models. This cycle repeats for R rounds.

In the conventional FL setup, the server and the clients train the same model architecture M, requiring similar computational resources for training. Instead, we consider an encoder—decoder architecture for vision tasks and relax this assumption by considering the scenario where the clients and the server share only a simple decoder module D. Formally, the server, with greater computational resources, maintains a large-scale model $M^s = D \circ O$ where O (Oracle) is a large-scale encoder. Clients, constrained by lower resources, use compact models $M^k = D^k \circ E$ where E is a lightweight encoder. The decoders are initialized as D, i.e., at the beginning of the federated optimization, $D^k = D \quad \forall k$, and then their optimization becomes the objective of the federated training. All models share the same decoder structure D, ensuring compatibility, while the encoders are different (i.e., $E \neq O$). Note that, as is common, the number of classes in each domain is assumed to be known to allow instantiation of the decoders.

Given an input image $\mathbf{X} \in \mathcal{X} \subset \mathbb{R}^{H \times W \times 3}$, the encoders extract feature representations $\mathbf{O} = O(\mathbf{X})$ and $\mathbf{F} = E(\mathbf{X})$, where $\mathbf{O}, \mathbf{F} \in \mathbb{R}^F$. Importantly, to enable seamless integration of client-side models with the shared decoder, the features extracted on the client-side must approximate those of the server, that is, $\mathbf{F} \simeq \mathbf{O}$ for any given input $\mathbf{X}$. To achieve this, we exploit a translator block at client-side and cross-architectural distillation during pretraining on samples from an auxiliary dataset $\mathcal{D}^s$ prior to federated optimization (Section IV-B1). The objective of the CA-FKT task is to enable the server to classify data from the same domain as the one privately attained by the clients $\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k$ without direct access to client data. To ensure privacy and efficiency, clients train only the decoder D, which is task-specific and operates on high-level representations [14], [15], [16]. In the multidomain scenario, after training, the domain-specific decoders D^i can be concatenated together on the server and attached to the Oracle encoder, enabling recognition in all domains with minimal performance loss (Section V).

IV. Proposed Method

A. FedPromo Framework

The framework is built on an encoder–decoder classification network. Its key innovation is supporting different encoders on the server and client sides while maintaining a common feature space, enabling a shared decoder across both environments. This approach bridges the computational gap between server and client. The server capitalizes on higher computing capacity with a computationally intensive, yet highly capable, encoder network. In contrast, at the client-side, with limited computational and energy resources, a significantly smaller and lighter network is used as a proxy for the server counterpart. In our experiments, we used DINOv2 [3] based on ViT-L/14-Reg [46] at the server side due to its ability to extract rich features across domains, while clients use a MobileNetV3-small [47], though any foundation and lightweight models can be used.

To allow cross-architectural distillation and enable decentralized training of a shared classifier, a translator block $T : \mathbb{R}^{F'} \mapsto \mathbb{R}^F$ that maps the feature space of the client encoder to that of the server is needed. In this setting $E = T \circ \hat{E}$, and $\hat{E} : \mathcal{X} \rightarrow \mathbb{R}^{F'}$. In our implementation, we use the concatenation of the DINOv2 class token with the global average pooling of patch tokens as the feature space [12], and we implement T as a linear transformation for efficiency. The feature dimensions are $F' = 576$ (MobileNet) and $F = 2048$ (DINOv2). However, any translator module suitable for the selected networks can be used. The translator is pretrained and remains frozen during federated optimization, ensuring consistent feature representations across clients and the server.

To minimize client-side computation, we use a single linear layer as the classifier, defined as $D : \mathbb{R}^F \rightarrow [0, 1]^C$, mapping the shared feature space to a discrete probability distribution over the C classes of the considered domain, where $C = |\mathcal{C}|$. Additionally, to regularize the gradients during federated training, we introduce an L2-normalization layer between the encoders and the decoder, leading to improved performance (see the *Supplementary Material*). For reference, in our setting, the client-side model (encoder + translator + classifier) has 100× fewer parameters than the server model. Fig. 2 illustrates our architecture and the interactions between server and clients.

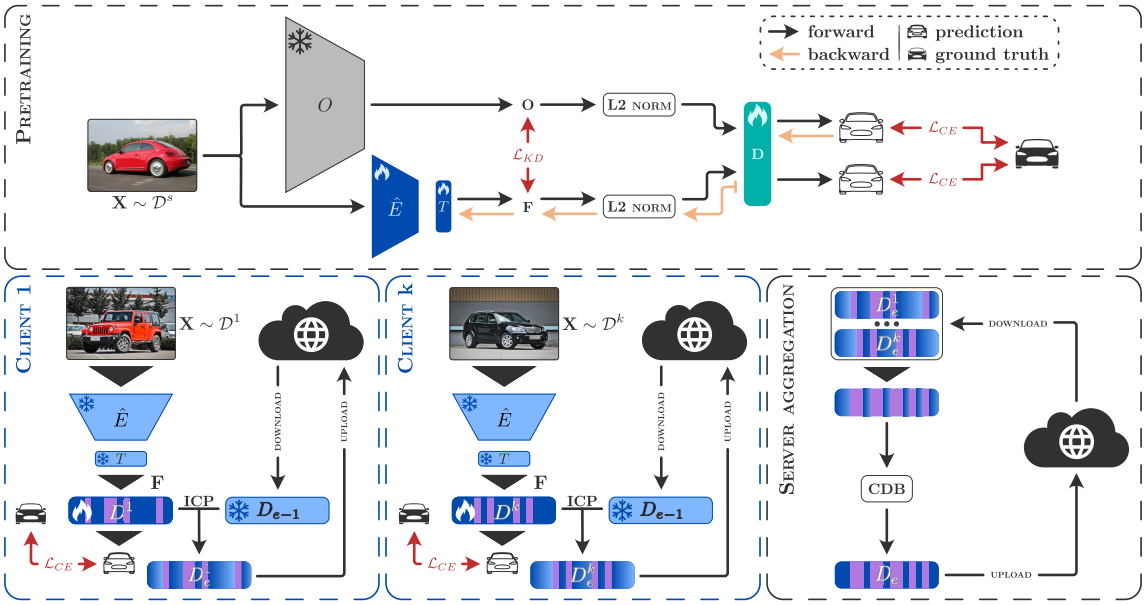


Fig. 2. FedPromo employs a two-stage architecture. The top row illustrates the cross-architectural distillation pretraining, where we transfer features useful for classification by training the decoder D only on oracle features, and keeping it frozen when the student feature vector $\mathbf{F}$ is in input. We also add an extra KD loss acting directly on the features. The bottom row summarizes the federated setup, depicting decentralized training on two sample clients and subsequent model aggregation on the server. Our inactive classes preservation (ICP) module reduces client divergence, while the class de-biasing (CDB) module acts as a distribution whitening heuristic for the decoders before they are sent to the server.

B. Training Procedure

FedPromo requires aligned feature spaces between server and client encoders to enable a shared decoder. We achieve this through cross-architectural distillation during pretraining, followed by federated training on private client data.

1) Cross-Architecture Pretraining: Our pretraining strategy couples the primary classification task with cross-architectural distillation [12]. The global optimization objective is given by $\mathcal{L} = \mathcal{L}_{KD} + \lambda \mathcal{L}_{CE}$, where $\mathcal{L}_{CE}$ is the categorical cross-entropy loss, and $\mathcal{L}_{KD}$ is the KD loss for feature reconstruction. We empirically set $\lambda = 0.5$. The $\mathcal{L}_{KD}$ term enforces feature alignment between student (client) and teacher (server) encoders by aligning both the norm and the orientation of the feature vectors in the high-dimensional embedding space. Formally, for a given training sample $\mathbf{X} \in \mathcal{D}^s$, the KD loss can be expressed as

$$\mathcal{L}_{KD} = \ell_1(\mathbf{F}, \mathbf{O}) + \ell_2(\mathbf{F}, \mathbf{O}) + \ell_{cos}(\mathbf{F}, \mathbf{O}) \tag{1}$$

where ℓ_1, ℓ_2 and ℓ_{cos} represent the corresponding distances.

For best performance, the dataset $\mathcal{D}^s$, where the cross-entropy loss is optimized, should be in-domain to the subsequent federated training data. To reinforce distillation, we train the classifier only on frozen teacher features and apply it directly to the student. This forces the student to replicate the most relevant features for classification according to the server model, improving accuracy (see Sections V and VI).

2) Federated Learning: The complexity of the CA-FKT scenario requires several novel contributions to the federated optimization process. We focus on regularizing the client-side training to maintain a high degree of alignment between client and server behaviors. This is crucial when considering a realistic IoT scenario where the federated partition is highly non-IID, resulting in minimal class overlap among clients.

We introduce two key innovations to address these issues.

1) A client-side regularization objective, ICP, which preserves knowledge of classes absent in client data that cross-entropy tends to diminish, inspired by continual learning [48], [49].
2) A weight filtering strategy, called CDB, is implemented at both client and server-side: it removes the average output activation for each class from the classifier, thereby reducing overall confusion.

See the *Supplementary Material* for a pseudocode of the implementation.

a) Inactive classes preservation: Our federated scenario also allows the learning of fine-grained datasets partitioned non-IID across IoT clients. This setup often leads to significant performance degradation when cross-entropy is the sole optimization loss: since each client k contains a limited number of classes, it can easily achieve a lower entropy score (and loss) by minimizing the predicted probability of classes absent from its set. Over time, this effect leads to overconfident predictions of frequent classes, even after aggregation.

We address this issue by implementing selective weight optimization for the clients. Essentially, each client keeps track of the labels encountered at each local epoch and enforces smaller updates on the classifier weights corresponding to the missing classes. Let us define D_e^k the decoder of client k at the end of current local epoch e (mapping features of size F to C classes), its weights as $\boldsymbol{\omega}_e \in \mathbb{R}^{C \times F}$, and $\mathcal{A} \subset \mathcal{C}$ the subset of active classes (i.e., those seen by the client). We can express the regularization strategy as

$$\boldsymbol{\omega}_e^{\text{ICP}}[c] = \begin{cases} \eta \boldsymbol{\omega}_e[c] + (1-\eta)\,\boldsymbol{\omega}_{e-1}[c], & \forall c \in \mathcal{C} \setminus \mathcal{A} \\ \boldsymbol{\omega}_e[c], & \forall c \in \mathcal{A} \end{cases} \tag{2}$$

where η has been empirically tuned to 0.05 (see *Supplementary Material* for details), and $\mathcal{C} \setminus \mathcal{A}$ is the set of inactive classes. Note that, at the beginning, we initialize the weights from the pretraining output model received from the server.

b) Class de-biasing: Our second contribution focuses on reducing common activations between classes in the classifier weights (i.e., feature whitening), which may leak local class distribution. Given the fine-grained nature of our setting, where classes often share many similarities (*e.g.*, all vehicles considered in CompCars are, in fact, cars), it is crucial to ensure that the classifier is not affected by the *bias* component common to all classes of the domain. This means that it should focus on distinguishing between the fine-grained classes, rather than categorizing them as their superclass. For example, in CompCars, it should focus on recognizing how each car differs from others, rather than on recognizing them all as cars. We perform de-biasing both locally at the clients and globally at the server, reducing crosstalk between classes and avoiding common feature activation from visually similar classes (see Section VI-F). Formally, before the aggregation step, each active client $k \in \mathcal{K}_A$ ($K_A = |\mathcal{K}_A|$) updates its classifier weights $\boldsymbol{\omega}_r^k = \boldsymbol{\omega}_{e_{\text{last}}}^{\text{ICP}}$ (e_{last} is the last local epoch) as

$$\boldsymbol{\omega}_{r+1}^k[c] = \boldsymbol{\omega}_r^k[c] - \frac{1}{C}\sum_{c=1}^{C} \boldsymbol{\omega}_r^k[c]. \tag{3}$$

This newly debiased weight is then sent to the server for aggregation, which performs the same operation on the weights $\boldsymbol{\omega}_{r+1}^s$ obtained after aggregation at the server-side.

V. Experimental Results

A. Implementation Details

We implemented our approach using the *Flower* framework [50] in PyTorch.[1] For pretraining, we train our architectures on the server for 60 epochs using a batch size of 64. We adopt a cosine-annealing learning rate scheduler (decaying to $lr = 0$) with a peak learning rate of $lr_{\max} = 0.005$. Optimization is performed using Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with no weight decay. We follow the TorchVision benchmark [51] for MobileNet data augmentation. For FL, we keep the optimizer, scheduler, batch size, and peak learning rate the same as in pretraining. The optimization runs for $R = 500$ rounds, with $e_{\max} = 10$ local epochs per client (without augmentation). We consider a federated setup with $K = 100$ clients, where $K_A = 10$ clients participate per round. The method-specific hyperparameters of the competitors are: FedMOON regularizer $\mu_{\text{MOON}} = 0.1$, FedMOON temperature $\tau = 1$, FedAVG + EMA decay $\eta = 0.05$, FedProx proximal term $\mu_{\text{Prox}} = 1$, FedHEAL scale $\beta = 0.4$. Pretraining requires 40 min on an NVIDIA L40s for StanfordCars and ~2 days for ImageNet; Federated finetuning on CompCars requires ~9 h for 500 rounds. See *Supplementary Material* for additional details on the other datasets. The document also reports preliminary results on real devices with communication cost analysis.

[1]Code is available at: `https://github.com/LTTM/FedPromo`

B. Datasets

We evaluate FedPromo on five domain pairs exhibiting long-tailed class distributions (average of 272 target classes, all belonging to similar objects) distributed on the clients according to a Dirichlet distribution with concentration parameter $\alpha = 1$ [52]. We include a representation of the employed splits in the *Supplementary Material*. In each scenario, a public source dataset is used for pretraining, while a private target dataset is distributed across clients in a federated setting. We consider two pairs of datasets containing images of animals (CUB200 [53] → NABirds [54], ImageNetPets → OxfordPets [55]), two pairs containing images of vehicles (StanfordCars [56] → CompCars [57], FGVCAircraft [58] → MilitaryAircraft [59]) and one pair containing images of food (Food101 [60] → UECFOOD256 [61]). The only custom dataset is *ImageNetPets*, which we extracted from the ImageNet-1k dataset [62] using the WordNet hierarchy [63] (we will release the split with the codebase, see the *Supplementary Material* for details). The source domain is considered public, allowing the server to pretrain a proxy model. This proxy model is then used by clients to learn knowledge from their private target domains in a federated manner. Note how the different couples of datasets have very different domain shifts, see the *Supplementary Material* for an analysis of this aspect. Results without using a pretraining dataset are also provided in the *Supplementary Material*.

C. Experimental Comparison

We compare FedPromo against six competitors using Top-1 and Top-5 classification accuracy. We evaluate for reference both the proxy models (MobileNetv3-s, 11.8 MB, 3M params, 0.123 GFLOPs) and the Oracle model (DINOv2/L14, 1.2 GB, 300M params, 158.2 GFLOPs), where the decoder trained in the federated setting is attached to the server encoder. The FLOPs values have been computed using the `ptflops` [64] library at a resolution of 224×224 pixels. First, we consider the baseline FedAvg [11] strategy and its extension with an exponential moving average, FedAvg + EMA (with rate 0.5). We then evaluate two benchmark federated strategies, FedProx [20] and MOON [23]. The former adds a regularization term to clients during training, named *proximal term*, slowing the evolution of the training but achieving greater stability. We consider the best-case scenario where no straggler clients are present in FedProx [20]. The latter extends FedProx with KD on network outputs and contrastive learning for feature alignment. Furthermore, we consider the recent FedBN [65] and FedHEAL [66] approaches; the former is a parameter-free approach that selectively aggregates the client parameters to account for distribution shifts, while the latter extends the FedAVG aggregation, introducing a similarity-based re-weighting factor. Finally, for an upper-bound reference, we report the *centralized* setup, where classifiers are trained using standard nonfederated optimization. Note that, similar to competitors, our centralized setup is meant to reflect the federated scenario: only the decoder is trained, and the features in input are those of the proxy model. Training runs for 50 epochs, matching the number of iterations in FL settings.

TABLE I

FEDERATED TOP-1 AND TOP-5 ACCURACY (↑) WHEN OPTIMIZATION STARTS FROM AN IN-DOMAIN DATASET.† PARTIALLY RE-IMPLEMENTED FOR CA-FKT. EVALUATION IS DONE ON THE SERVER-SIDE AFTER FEDERATED AGGREGATION. BEST IN BOLD, SECOND-BEST UNDERLINED

$\mathcal{D}$	Method	MobileNet		DINOv2	
		Top-1	Top-5	Top-1	Top-5
CompCars	FedAvg	26.4	43.4	14.0	27.4
	FedAvg + EMA	18.5	35.9	21.5	42.6
	FedProx†	9.5	24.2	13.7	31.2
	MOON†	26.4	43.3	13.8	27.4
	FedBN†	26.4	43.4	14.0	27.4
	FedHEAL†	26.3	43.3	13.9	27.0
	FedPromo (ours)	**37.2**	**63.5**	**35.8**	**69.4**
	Centralized	48.3	75.9	41.1	73.2
UECFOOD256	FedAvg	48.6	74.4	29.6	50.4
	FedAvg + EMA	41.4	73.0	49.1	83.1
	FedProx†	5.6	30.1	8.2	42.4
	MOON†	48.4	74.3	29.7	50.5
	FedBN†	48.6	74.4	29.6	50.4
	FedHEAL†	48.5	74.2	29.5	50.2
	FedPromo (ours)	**52.1**	**80.0**	**58.7**	**88.7**
	Centralized	55.3	83.0	60.1	88.3
NABirds	FedAvg	23.6	41.2	6.1	12.9
	FedAvg + EMA	20.7	41.1	20.7	43.6
	FedProx†	12.8	31.4	16.0	39.9
	MOON†	23.8	41.4	6.1	12.9
	FedBN†	23.6	41.2	6.1	12.9
	FedHEAL†	22.1	39.4	6.3	13.1
	FedPromo (ours)	**32.2**	**59.0**	**30.3**	**65.0**
	Centralized	35.3	62.8	32.3	65.3
MilitaryAircraft	FedAvg	10.7	29.3	9.2	27.7
	FedAvg + EMA	8.1	26.7	**12.4**	**39.0**
	FedProx†	6.9	23.5	10.4	33.8
	MOON†	**10.8**	29.1	9.6	27.8
	FedBN†	10.7	29.3	9.2	27.7
	FedHEAL†	**10.8**	**29.7**	9.5	27.9
	FedPromo (ours)	9.9	29.3	11.9	32.1
	Centralized	11.3	31.5	13.4	35.4
OxfordPets	FedAvg	56.9	89.1	47.9	79.2
	FedAvg + EMA	52.4	87.7	67.6	96.4
	FedProx†	54.1	89.1	68.0	95.0
	MOON†	56.5	88.8	47.4	78.8
	FedBN†	56.9	89.1	47.9	79.2
	FedHEAL†	56.6	88.6	50.1	79.5
	FedPromo (ours)	**59.2**	**91.6**	**72.7**	**98.0**
	Centralized	61.6	92.2	71.2	94.4

Comparison with federated personalization techniques is also discussed in the *Supplementary Material*.

D. Results With In-Domain Pretraining

Table I shows the strong and consistent performance of FedPromo across various scenarios. Our approach outperforms existing methods in most cases, achieving the highest Top-1 and Top-5 accuracy in eight out of ten scenarios, while ranking among close to the best in the remaining cases, demonstrating its robustness.

In particular, improvements over existing methods are often substantial. For example, in the StanfordCars→CompCars setup, FedPromo achieves a Top-1 accuracy of 37.2% for the proxy models, outperforming the next best method by 10.8 percentage points. Similarly, in the Food101 →UECFOOD256 scenario, our approach reaches 58.7% Top-1 accuracy when applied to DINOv2 (*server* model), outperforming the second-best method (FedAvg + EMA) by 9.6 percentage points. The gain is also evident in the CUB200 →NABirds setting, where FedPromo achieves a server Top-1 accuracy of 30.3%, compared to 16.0% of FedProx and 20.7% of FedAvg + EMA. These results highlight FedPromo's advantage over competing techniques, particularly when leveraging the server-side DINOv2 model. A similar improvement is also notable in the ImageNetPets → OxfordPets setting, where our approach is consistently the best performer, achieving an average accuracy improvement of 2.8%. Throughout all settings, FedPromo's performance is very consistent across random seeds, with a standard deviation never exceeding 0.5 (average of 0.16), see Table S.12 in the *Supplementary Material* for more details.

A key strength of FedPromo in the CA-FKT task is its consistency across different architectures. Our approach delivers top-tier performance for both MobileNet (proxy) and DINOv2 (server), demonstrating: 1) versatility across different network designs and 2) robust feature alignment through effective pretraining. Even in the most challenging case (FGVCAircraft→MilitaryAircraft), FedPromo remains on par with the best competitors, ensuring reliable performance across diverse domains. Compared to the upper-bound (centralized training), FedPromo shows significant promise. On proxy models, it achieves an average gap of just 4.8% (Top-1) and 5.1% (Top-5). On server models, this difference shrinks to 0.3% (Top-1) and 4.9% (Top-5), approaching the upper-bound.

Our experiments suggest that an auxiliary dataset should have DINO cosine similarity ≥ 0.5 to the target domain for reliable performance (refer to S.12 for details).

E. Results With Out-of-Domain Pretraining

To further assess the generalization of FedPromo, we evaluate its performance using ImageNet-1k [62] for pretraining, without any in-domain pretraining dataset (see Table II). FedPromo remains highly effective even without in-domain pretraining, reinforcing its adaptability. It achieves performance close to centralized training, both on the proxy model and the server model, outperforming competitors on both Top-1 and Top-5 in eight out of ten settings, getting very close to the best in the remaining ones. In particular, performance gains are substantial on the *NABirds*, *UECFOOD*, and *CompCars* datasets, while the MilitaryAircraft is challenging for all approaches, and performance on OxfordPets tends to saturate, making it difficult to see the difference since ImageNet contains many of the target classes.

F. Multidomain Results

Table III reports the accuracy achieved by the DINOv2 model when classifying data from all the domains together after distributed optimization using FedPromo. We compare

TABLE II

EXPERIMENTAL RESULTS WHEN FEDERATED OPTIMIZATION STARTS FROM AN OUT-OF-DOMAIN DATASET. FEDERATED TOP-1 AND TOP-5 ACCURACY (↑) STARTING FROM IMAGENET-1K (1000 CLASSES) PRETRAINING. † PARTIALLY RE-IMPLEMENTED FOR CA-FKT. BEST IN BOLD, SECOND-BEST UNDERLINED

$\mathcal{D}$	Method	MobileNet Top-1	MobileNet Top-5	DINOv2 Top-1	DINOv2 Top-5
CompCars	FedAvg	16.2	31.4	1.2	3.9
	FedAvg + EMA	6.5	17.5	3.0	9.7
	FedProx†	2.1	7.6	2.4	8.5
	MOON†	16.2	31.5	1.3	3.9
	FedBN†	16.2	31.4	1.2	3.9
	FedHEAL†	16.3	31.5	1.2	4.0
	FedPromo (ours)	**18.3**	**37.1**	**3.3**	**10.8**
	Centralized	29.7	53.1	3.4	9.5
UECFOOD256	FedAvg	43.2	68.9	2.1	6.9
	FedAvg + EMA	36.2	64.5	23.5	48.4
	FedProx†	3.0	13.0	4.0	13.2
	MOON†	43.8	68.4	2.0	7.1
	FedBN†	43.2	68.9	2.1	6.9
	FedHEAL†	43.9	68.8	2.1	7.0
	FedPromo (ours)	**48.2**	**75.8**	**24.0**	**48.5**
	Centralized	51.8	78.4	27.0	56.5
NABirds	FedAvg	28.0	46.7	1.5	5.1
	FedAvg + EMA	21.2	41.8	11.3	29.1
	FedProx†	10.3	25.7	10.5	26.6
	MOON†	28.0	46.5	1.6	5.0
	FedBN†	28.0	46.7	1.5	5.1
	FedHEAL†	28.0	46.6	1.4	4.9
	FedPromo (ours)	**37.5**	**65.1**	**18.5**	**48.6**
	Centralized	42.7	70.5	17.3	47.6
MilitaryAircraft	FedAvg	16.4	38.1	5.1	16.2
	FedAvg + EMA	10.6	28.9	8.0	**25.8**
	FedProx†	8.8	24.2	**8.3**	24.8
	MOON†	16.5	38.2	5.3	16.2
	FedBN†	16.4	38.1	5.1	16.2
	FedHEAL†	16.4	38.2	5.1	16.5
	FedPromo (ours)	**16.9**	**40.0**	7.1	24.1
	Centralized	18.9	43.4	6.7	25.9
OxfordPets	FedAvg	**82.4**	97.2	84.0	99.5
	FedAvg + EMA	77.1	97.4	85.2	99.8
	FedProx†	78.2	97.4	85.6	**99.9**
	MOON†	**82.4**	97.4	84.7	99.5
	FedBN†	**82.4**	97.2	84.0	99.5
	FedHEAL†	**82.4**	97.3	84.5	99.5
	FedPromo (ours)	82.2	**98.2**	**89.5**	99.8
	Centralized	83.7	98.2	87.8	99.8

the domain-*specific* accuracy (same configuration as Table I) with that of the domain-*agnostic* approach, where we concatenate all the classifiers of the single domains into a single larger one. In the latter case, our method achieves competitive results, closely matching the domain-specific performance. The average loss of top-1 accuracy across domains is just 1.9%, while for top-5 the accuracy loss is 5.8%. Note that most of the performance degradation arises from the MilitaryAircraft dataset, where low starting accuracy leads to confusion with classes in other domains when concatenating the classifiers. More details are in the *Supplementary Material*.

TABLE III

EXPERIMENTS ON MULTIDOMAIN SCENARIO, SERVER ACCURACY. SPECIFIC: CLASSIFIERS ARE KEPT SEPARATED (REQUIRES SAMPLES' DOMAIN KNOWLEDGE). AGNOSTIC: CLASSIFIERS ARE CONCATENATED, UNKNOWN SAMPLES (NO DOMAIN INFORMATION ASSUMED)

Dataset	Specific Top-1	Specific Top-5	Agnostic Top-1	Agnostic Top-5
CompCars	35.8	69.4	35.8	69.1
UECFOOD256	58.7	88.7	58.7	88.4
NABirds	30.3	65.0	30.2	64.8
Military Aircraft	11.9	32.1	4.6	8.9
OxfordPets	72.7	98.0	70.9	92.8
Average	41.9	70.6	40.0	64.8

TABLE IV

METHOD COMPONENTS ABLATION. FEDERATED TOP-1 AND TOP-5 ACCURACY (↑) ON THE COMPCARS (431 CLASSES) DATASET STARTING FROM STANFORDCARS (196 CLASSES) PRETRAINING

CDB	ICP	MobileNet Top-1	MobileNet Top-5	DINOv2 Top-1	DINOv2 Top-5	Avg.
✗	✗	26.4	43.4	14.0	27.4	27.8
✓	✗	26.4	43.4	13.9	27.5	27.8
✗	✓	36.8	63.3	35.6	69.2	51.2
✓	✓	37.2	63.5	35.8	69.4	51.5

VI. ABLATION STUDY

In this section, we present our ablation study, analyzing the contribution of each component, and discussing some limitations of our approach. Here we report experiments in the StanfordCars→CompCars scenario. Ablation results on all other domains are available in the *Supplementary Material*.

A. Method Components' Ablation

Table IV presents the results of our component ablation study on the CompCars dataset, in which we investigate the effect of the CDB and ICP modules in our federated approach. The baseline performance without both of them is relatively low for both proxy and server models. Adding the ICP module results in a significant accuracy boost. CDB alone has a limited effect, but adding it on top of ICP is more effective in achieving the best performance in all cases, in spite of the minimal computational cost. These results confirm that both components effectively contribute to our approach. The ablation data on the other four datasets are in the *Supplementary Material* (Tables S7–S10), which confirms the findings presented here for the CompCars experiments.

B. Feature Alignment

Figs. 3 and 4 show t-SNE plots of the feature alignment achieved by our pretraining strategy. More specifically, the first shows the embedding of the features before distillation, while the second shows them after. The plot clearly illustrates how the features of DINO (○) and MobileNet (□) are well-aligned class-by-class and clustered in the same regions. Furthermore, the DINO class prototypes (☆) are located close to their respective feature clusters, further confirming the alignment. For better clarity, the figures present only a subset of classes. A full visualization is available in the *Supplementary Material* (Figs. S3-S4). These results highlight the pretraining strategy's

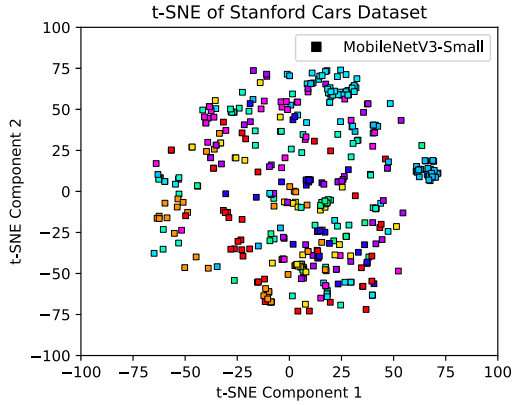


Fig. 3. t-SNE plot showing the alignment of the features before pretraining. For clarity, it shows the embeddings of just ten classes. The full plot with all classes is in the *Supplementary Material*.

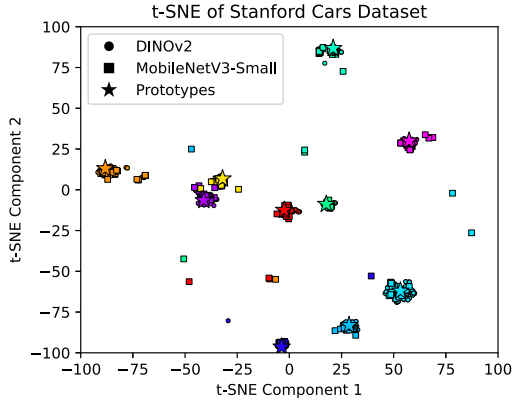


Fig. 4. t-SNE plot showing the alignment of the features after pretraining. For clarity, it shows the embeddings of just ten classes. The full plot with all classes is in the *Supplementary Material*.

effectiveness in aligning features across models, which is critical for federated knowledge transfer.

C. Differential Privacy

To study resilience against privacy attacks, we used the (ε, δ)-differential privacy framework [67], [68]. More details are reported in the *Supplementary Material* Table V reports the experiments on the CompCars dataset, where we fix $\delta = 1e{-}5$ and vary ε (the other parameters are as in the Flower [50] framework). FedPromo performs well even at very low values of ε, which correspond to a very high overall privacy budget. Indeed, even at $\varepsilon = 5$, the average accuracy is 46.7%, that is just 4.8% less than the setting without DP.

D. Different Federated Training Configurations

To further analyze the stability of the training, we performed experiments varying the number of rounds, total and active clients, the probability of transmission failure, and the Dirichlet parameter α. Detailed results for these settings, comparing FedPromo with the FedAvg baseline, are reported in the *Supplementary Material*. Tests on the number of rounds show that $R = 100$ rounds are insufficient for convergence,

TABLE V

DIFFERENTIAL PRIVACY [67] EXPERIMENTS VARYING ε WITH $\delta = 10^{-5}$. MORE DETAILS IN THE SUPPLEMENTARY MATERIAL

ε	Client		Server	
	Acc-1	Acc5	Acc-1	Acc-5
✗	37.2	63.5	35.8	69.4
50	35.3	61.5	34.7	67.8
10	34.9	61.1	33.6	66.6
5	33.7	59.8	31.0	62.4
2.5	30.4	56.4	24.7	54.0
1	19.8	42.0	14.2	33.7

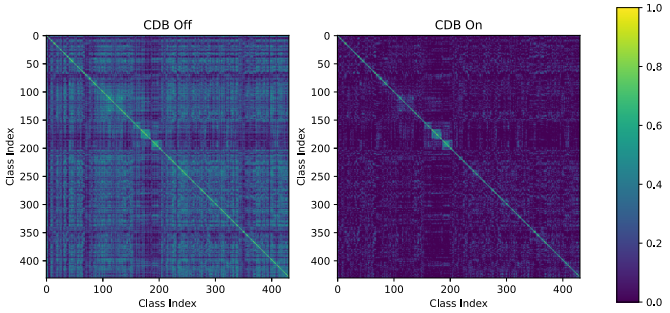


Fig. 5. Self-similarity plots of the CompCars classifier weights trained with CDB off (left) and with CDB on (right). Our CDB block drastically reduces crosstalk between classes.

leading to reduced accuracy. On the other hand, from 250 rounds and above, the average accuracy stabilizes near 51.1, with the highest value corresponding to $R = 500$. With fewer total clients ($K = 50$), the average accuracy improved by 4.1%, while with more clients ($K = 150$ and 300), it decreased by 9.4% and 15.3%, respectively, due to increased data heterogeneity and reduced class coverage on each client in this more challenging setting. Similarly, decreasing the number of active clients per round to $K_A = 5$ improves accuracy by 1.8%, while increasing it to $K_A = 20$ reduces it by 2.1%. To assess robustness, we simulated random uplink (client→server) transmission dropout with probabilities 10%, 30%, and 50%, observing an average accuracy degradation of ~0.1% in all cases. Finally, varying the Dirichlet parameter, we observe that increasing α to 10 leads to an average improvement of 2.5%, while further increases in the parameter value lead to negligible variations. Finally, note how in all these settings our approach consistently outperforms FedAVG by a large margin. Additional experiments with a larger trainable head on the clients (i.e., adding to the full encoder a 3-layer FCN) are reported in Section S.13 of the *Supplementary Material*. They show that the added extra complexity does not bring relevant performance improvements.

E. Strength of ICP Regularization

Table VI reports a study on the parameter η that controls the update of inactive classes in the ICP block. Completely disabling updates on inactive classes ($\eta = 0$) results in subpar performance. On the other side, using a large value ($\eta = 0.5$) causes an even stronger accuracy degradation. We experimentally found that setting $\eta = 0.05$ provides the highest overall accuracy.

TABLE VI

TUNING OF η. WE FOCUS ON THE BEST OVERALL PERFORMANCE. FEDERATED TOP-1 AND TOP-5 ACCURACY (↑) ON COMPCARS (431 CLASSES) STARTING FROM STANFORDCARS (196 CLASSES)

η	MobileNet		DINOv2		Avg.
	Top-1	Top-5	Top-1	Top-5	
0.00	34.9	61.2	35.1	68.9	50.0
0.05	37.0	63.4	36.1	69.5	51.5
0.10	38.1	63.7	35.4	67.4	51.2

F. Effect of CDB

Fig. 5 illustrates the impact of the CDB technique on the classifier weights. We start by normalizing the classifier parameters $\boldsymbol{\omega} = \boldsymbol{\omega}_R^s / \|\boldsymbol{\omega}_R^s\|_2$ across the feature dimension, before computing and plotting the similarity matrix $\mathbf{S} = \boldsymbol{\omega}\,\boldsymbol{\omega}^\intercal \in \mathbb{R}^{C\times C}$. This matrix captures the similarity between each class embedding and all the others. High off-diagonal activations indicate crosstalk between classes, leading to low-confidence predictions, since a single feature will activate multiple classes at the same time. CDB minimizes class bias by pushing the self-similarity matrix closer to an identity matrix, reducing cross-class interference. In the figure, we compare the matrices computed on the last aggregated models trained with the CDB block disabled (left) and with CDB turned on (right). We can appreciate how disabling the block leads to significant crosstalk and confused predictions, while the full framework (thanks to CDB) substantially reduces bias, resulting in sharper, more confident classifications.

VII. CONCLUSION

We introduced FedPromo, a novel FL framework to adapt a large server-side FM to local client data without accessing it, thereby preserving privacy. This approach is particularly suitable for IoT scenarios, where efficiency and bandwidth constraints are critical. Our method pretrains a lightweight proxy model to generate features aligned with the FM, significantly reducing computational demands on resource-constrained devices. The proxy model trains a classifier at remote clients in a federated setting, allowing seamless knowledge transfer while minimizing energy consumption and bandwidth usage through a smaller shared model. Once trained, the classifier is attached to the FM, improving its generalization without direct access to client data. Experimental results demonstrate FedPromo's effectiveness on multiple datasets, showcasing its potential. FedPromo's compute efficiency makes it promising for applications in smart cities, industrial IoT, and environmental monitoring systems, where distributed devices need to collaborate while preserving privacy and minimizing resource usage. Future work will focus on improving performance and extending the approach to more complex scene understanding tasks, such as object detection or semantic segmentation.

VIII. LIMITATIONS

FedPromo enables efficient and privacy-aware adaptation of FMs in federated settings. The method relies on access to a *task-specific* public dataset to achieve the best performance, which may not always be available. Performance can vary between domains, with weaker results in highly heterogeneous or challenging scenarios (*e.g.*, MilitaryAircraft dataset). FedPromo is currently limited to classification tasks, and its extension to more complex settings, such as object detection or segmentation, requires additional components. Moreover, the current multidomain experiment covers five domains with disjoint class-sets, and may not extend as-is to overlapping scenarios. Ethically, while raw data remains on client devices, shared model components may still leak information if subject to attacks, highlighting the need for privacy-preserving and class-regularization techniques.

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Open Access funding provided by 'Universita degli Studi di Padova' within the CRUI CARE Agreement